

SECSR2043 OPERATING SYSTEMS

[20 Marks]

Name : Evelyn Goh Yuan Qi
 Student : A23CS0222
 ID :
 Section : 2

Marks

Instruction: Please answer all of the following questions. Whenever the 🙋 symbol appears, please raise your hand to call your instructor, he/she will verify your results by putting his / her initial next to the symbol.

1. Type the following commands using a text editor and save it as a *yourname.sh* (Example: ahmad.sh).

```
echo "Hello world" > helloworld.jar
mkdir cars; mkdir dates; mkdir fruits drinks
cd cars; echo "Honda Accord" > accord.c
cp accord.c civic.c; echo proton > proton.c; cd ../dates;
date > dateoftheday
cat dateoftheday > appointment
cd ../fruits; echo apple > apple.txt; cat apple.txt >
orange.txt
cd drinks; cp ../cars/*.sh .; cp ../fruits/*.sh .;
cp ../*.jar .
```

- a) Execute the script and draw a tree structure that contains created directories and files. The parent node of the directory begin with **\$HOME** directory.

[4 marks]



Print screen the script that you type;

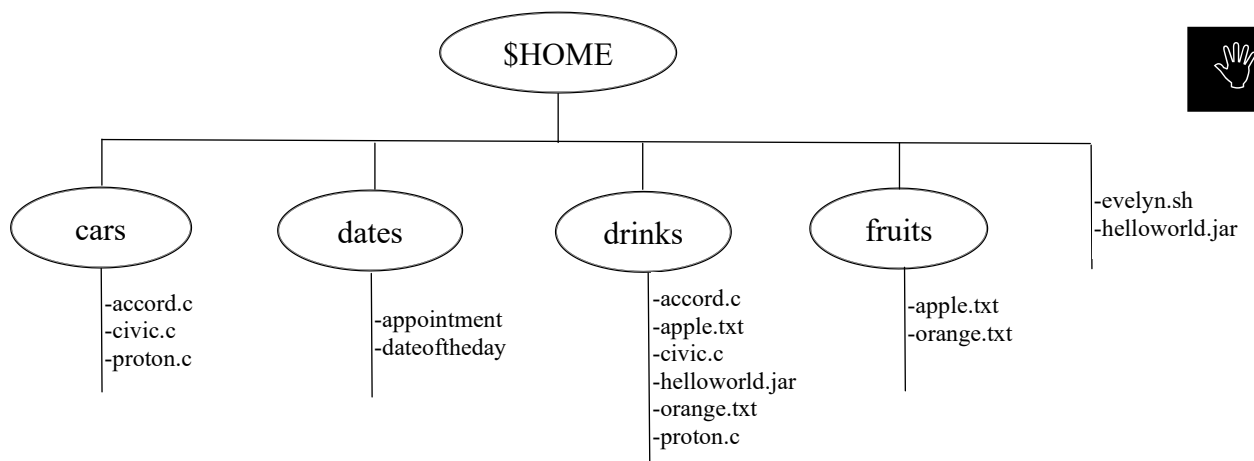
```
evelyn@secr2043:~$ nano evelyn.sh
```

```
GNU nano 7.2 evelyn.sh
echo "Hello world" > helloworld.jar
mkdir cars; mkdir dates; mkdir fruits drinks
cd cars; echo "Honda Accord" > accord.c
cp accord.c civic.c; echo proton > proton.c; cd ../dates;
date > dateoftheday
cat dateoftheday > appointment
cd ../fruits; echo apple > apple.txt; cat apple.txt > orange.txt
cd drinks; cp ../cars/*.c .; cp ../fruits/*.c .;
cp ../*.jar .
```



Then draw the tree

```
evelyn@secr2043:~$ tree
.
|-- cars
|   |-- accord.c
|   |-- civic.c
|   `-- proton.c
|-- dates
|   |-- appointment
|   `-- dateoftheday
|-- drinks
|   |-- accord.c
|   |-- apple.txt
|   |-- civic.c
|   |-- helloworld.jar
|   |-- orange.txt
|   `-- proton.c
|-- evelyn.sh
|-- fruits
|   |-- apple.txt
|   `-- orange.txt
`-- helloworld.jar
```



- b) Write an interactive bash script that will read a type of file extension, display all those files, and count the number of files. To validate your script, display c program files, and enter “c” as the input to the bash script. [4 marks]

Print screen the bash script you type and run

```
GNU nano 7.2 filefinder.sh *
#!/bin/bash

# Prompt for file extension
read -p "Enter file extension (e.g., c, txt): " ext

# Find and display matching files
echo "Files with .$ext extension:"
find . -type f -name ".*$ext"

# Count matching files
count=$(find . -type f -name ".*$ext" | wc -l)
echo "Number of .$ext files: $count"
```

```
evelyn@secr2043:~$ nano filefinder.sh
evelyn@secr2043:~$ chmod +x filefinder.sh
evelyn@secr2043:~$ ./filefinder.sh
Enter file extension (e.g., c, txt): c
Files with .c extension:
./drinks/civic.c
./drinks/accord.c
./drinks/proton.c
./cars/civic.c
./cars/accord.c
./cars/proton.c
Number of .c files: 6
```



2. The following Figure 1 illustrates a tree structure of some directories and files.

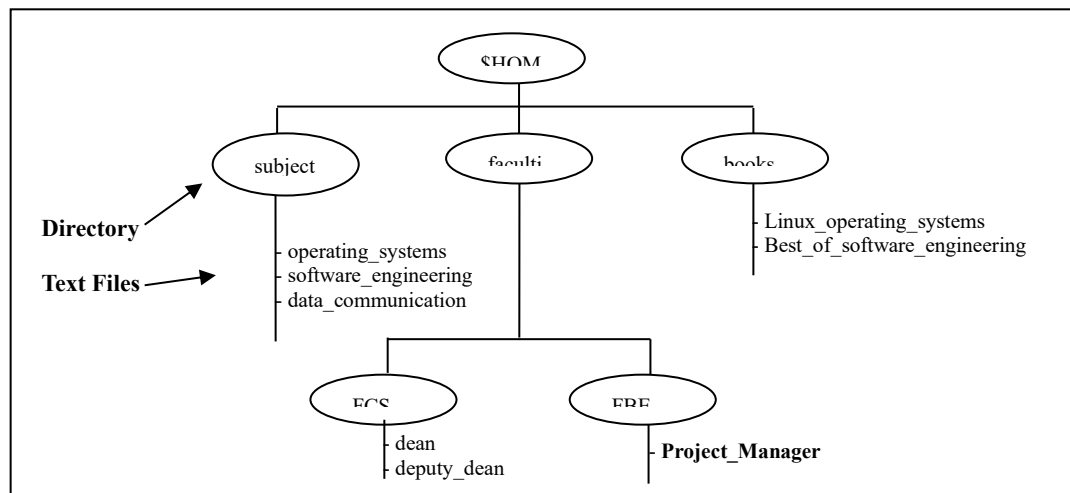


Figure 1

- a) Write a bash script (called `myname2a.sh`) that will produce directories and files as in Figure 1. Each text files contain its filename without the underscore character. For example: text file `Project_Manager` contains `Project Manager`). [4 marks]



Print screen the bash script you type and run

```
GNU nano 7.2 evelyn2a.sh *
#!/bin/bash

# Create directories
mkdir -p $HOME/subjects
mkdir -p $HOME/faculties/FCS
mkdir -p $HOME/faculties/FBE
mkdir -p $HOME/books

# Create and write to text files in the subjects directory
echo "operating systems" > $HOME/subjects/operating_systems
echo "software engineering" > $HOME/subjects/software_engineering
echo "data communication" > $HOME/subjects/data_communication

# Create and write to text files in the faculties/FCS directory
echo "dean" > $HOME/faculties/FCS/dean
echo "deputy dean" > $HOME/faculties/FCS/deputy_dean

# Create and write to text files in the faculties/FBE directory
echo "Project Manager" > $HOME/faculties/FBE/Project_Manager

# Create and write to text files in the books directory
echo "linux operating systems" > $HOME/books/linux_operating_systems
echo "Best of software engineering" > $HOME/books/Best_of_software_engineering
```

```
evelyn@secl2043:~$ tree $HOME
/home/evelyn
|-- books
|   |-- Best_of_software_engineering
|   `-- linux_operating_systems
|-- cars
|   |-- accord.c
|   |-- civic.c
|   `-- proton.c
|-- dates
|   |-- appointment
|   `-- dateoftheday
|-- drinks
|   |-- accord.c
|   |-- apple.txt
|   |-- civic.c
|   |-- helloworld.jar
|   |-- orange.txt
|   `-- proton.c
|-- evelyn.sh
|-- evelyn2a.sh
|-- faculties
|   |-- FBE
|   |   |-- Project_Manager
|   `-- FCS
|       |-- dean
|       `-- deputy_dean
|-- filefinder.sh
|-- fruits
|   |-- apple.txt
|   `-- orange.txt
|-- helloworld.jar
`-- subjects
    |-- data_communication
    |-- operating_systems
    `-- software_engineering

10 directories, 25 files
```

- b) Complete the following table by writing the access control of directories or files that were produced. Given is the access control for directory called book.

[2 marks]

Directory/File	Access Control
books	drwxr-xr-x
subjects	drwxr-xr-x
Best_of_software_engineering	-rw-r--r--
FCS	drwxr-xr-x
project_manager	-rw-r--r--



```
evelyn@secur2043:~$ ls -ld $HOME/subjects
drwxr-xr-x 2 evelyn evelyn 4096 Jun 14 02:55 /home/evelyn/subjects
evelyn@secur2043:~$ ls -l $HOME/books/Best_of_software_engineering
-rw-r--r-- 1 evelyn evelyn 29 Jun 14 02:55 /home/evelyn/books/Best_of_software_engineering
evelyn@secur2043:~$ ls -ld $HOME/faculties/FCS
drwxr-xr-x 2 evelyn evelyn 4096 Jun 14 02:55 /home/evelyn/faculties/FCS
evelyn@secur2043:~$ ls -l $HOME/faculties/FBE/Project_Manager
-rw-r--r-- 1 evelyn evelyn 16 Jun 14 02:55 /home/evelyn/faculties/FBE/Project_Manager
evelyn@secur2043:~$ ls -ld $HOME/books
drwxr-xr-x 2 evelyn evelyn 4096 Jun 14 02:55 /home/evelyn/books
```

- c) Write another bash script (called myname2c.sh) that will change the access control of the directories and files based on the following information:

[4 marks]

Directory/File	Users								
	Owner			Group			Public		
subjects	✓	✓	✓	✓	x	x	✓	x	x
Best_of_software_engineering	✓	x	✓	x	✓	x	x	x	x
FCS	✓	✓	x	x	x	x	✓	✓	✓
project_manager	x	x	x	x	✓	✓	x	x	✓

Print screen the bash script you type and run

```

GNU nano 7.2 evelyn2c.sh *
#!/bin/bash

# Set permission for 'subjects'
chmod 744 $HOME/subjects

# Set permission for 'Best_of_software_engineering'
chmod 520 $HOME/books/Best_of_software_engineering

# Set permission for 'FCS'
chmod 607 $HOME/faculties/FCS

# Set permission for 'Project_Manager'
chmod 031 $HOME/faculties/FBE/Project_Manager

evelyn@secur2043:~$ nano evelyn2c.sh
evelyn@secur2043:~$ chmod +x evelyn2c.sh
evelyn@secur2043:~$ ./evelyn2c.sh
evelyn@secur2043:~$ ls -ld $HOME/subjects
drwxr--r-- 2 evelyn evelyn 4096 Jun 14 02:55 /home/evelyn/subjects
evelyn@secur2043:~$ ls -l $HOME/books/Best_of_software_engineering
-r-x-w---- 1 evelyn evelyn 29 Jun 14 02:55 /home/evelyn/books/Best_of_software_engineering
evelyn@secur2043:~$ ls -ld $HOME/faculties/FCS
drw----rwx 2 evelyn evelyn 4096 Jun 14 02:55 /home/evelyn/faculties/FCS
evelyn@secur2043:~$ ls -l $HOME/faculties/FBE/Project_Manager
-----wx--x 1 evelyn evelyn 16 Jun 14 02:55 /home/evelyn/faculties/FBE/Project_Manager

```

- d) Complete the following table by writing the access control for each directory or file after executing the bash script in question 2(c)). [2 marks]



Directory/File	Access Control
subjects	drwxr--r--
Best_of_software_engineering	-r-x-w----
FCS	drw----rwx
project_manager	-----wx--x

End of Lab 3

*** All the Best for Final Exam ***